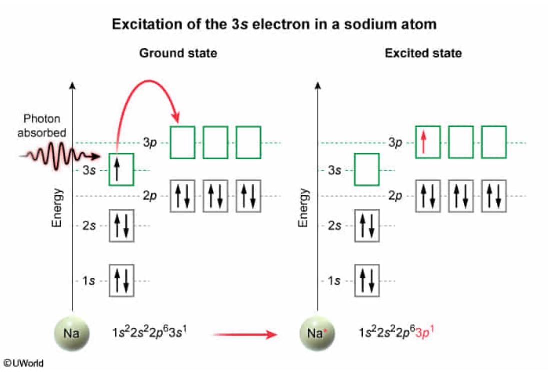
A sodium atom in an excited state has an electron configuration of $1s^2 2s^2 2p^6 3p^1$. The excited electron within the configuration was excited from the:

- A. 2*s* orbital. [2%]
- B. 2*p* orbital. [9%]
- C. 3*s* orbital. [70%]
- D. 3*p* orbital. [17%]

Correct
70% answered correctly

Explanation:



Electrons around an atom can exist only in particular, discrete energy levels, which are identified by quantum numbers and by orbital designations. Electron configurations account for the locations of all electrons within the energy levels (orbital shells and subshells) of an atom or ion. Shells are indicated using the principal quantum number *n*, which relates to the distance of an orbital from the nucleus. Subshells are labeled by type (*s*, *p*, *d*, *f*), which relates to the shapes of the orbitals that hold a maximum of 2, 6, 10, and 14 electrons, respectively.

Electron configurations are listed sequentially in order of increasing energy. In the **ground state** (unexcited state), the electrons of an atom occupy the lowest energy configuration. However, an electron in the ground state can temporarily achieve an excited state and jump from a **lower energy orbital** to a vacant **higher energy orbital** by **absorbing energy** equal to the **energy difference** between two levels. Upon relaxation back to the ground state, an excited electron releases the absorbed energy as a photon.

According to the filling order, the lowest energy configuration (ground state) of a sodium atom is $1s^2 2s^2 2p^6 3s^1$. As such, an electron configuration of $1s^2 2s^2 2p^6 3p^1$ for a sodium atom in an excited state indicates that the excited atom has a vacant 3*s* orbital (absent in the configuration) and an excited electron in the 3*p* orbital. Therefore, the excited electron occupying the 3*p* orbital was excited from the 3*s* orbital.

(Choices A and B) The 2*s* and 2*p* orbitals are fully occupied. Therefore, no electrons were excited from these orbitals in the ground state.

(Choice D) The 3*p* orbital is unoccupied in the ground state and cannot be the orbital from which the electron is excited.

Educational objective:
Electrons typically occupy the lowest energy configuration (ground state), but an electron may achieve an excited state and temporarily jump to a higher energy orbital by absorbing energy equal to the energy difference between the two levels.

Time spent: 0 sec	**QID**:402687
Subject:	**System:**
General Chemistry	Atomic Theory and Chemical Composition